at the 5% level. Probabilities predicted from this model are comparable between genders (see Online Appendix Figures O-A12–O-A15), with the first spell estimates showing greater gender similarity than the estimates at the 7^{th} year.[47]

We note that the parameters associated with publication in non-T5 journals and the X that are used in constructing these predictions are not allowed to vary by gender. Therefore, any differences in predicted probabilities stem from gender differences in tenure rates that are unrelated to differences in rewards associated with publication. Unlike the gender-specific publication rewards estimated in the duration analysis of Section 2.4, these logit estimates do not show any differences in rewards to publication by gender. It is not possible to estimate more sophisticated gender-specific publication specifications due to limited sample sizes for women.[48]

A recent study by Sarsons (2017) suggests the possible existence of bias in favor of male faculty in the tenure evaluation process. Specifically, her study finds that female faculty reap lower rewards (i.e., increases in the probability of receiving tenure) for papers co-authored with male co-authors compared to papers co-authored with female co-authors or solo-authored papers. The author observes that such patterns could arise if tenure committees are biased in favor of males in their attribution of credit for co-authored work. A competing hypothesis is that the nature of co-author pairings between males and females could differ from other types of pairings in a way that generates the pattern observed in the data. Ductor et al. (2018) document differences in co-author characteristics by authors' gender. The authors find that that co-author networks exhibit gender homophily. Further, they find that female authors tend to collaborate with smaller groups of co-authors who are more likely to be inter-linked with one another. Lastly, they find that females tend to collaborate with more senior faculty.

[47] These probabilities are obtained by adding a gender indicator variable to prediction Equation TA-2 to obtain $Pr(Tenure = 1 \mid \#\widehat{J} = \widehat{N}, \#\widetilde{J} = 0, \text{Gender} = g, \boldsymbol{X})$. The parameters used in these predictions are obtained by estimating Equation TA-1

[48] Many of the publication parameters are non-estimable for females due to sample size issues. Females account for only approximately 20% of the sample.

Our paper does not address the larger question of female representation in academic economics. A growing body of literature discusses female representation within the discipline and across academia in general (especially in the sciences). Analyzing data from the Survey of Earned Doctorates for the period 1974–2000, Ginther and Kahn (2004) find that women account for a substantially smaller share of doctorate degrees in economics and the physical sciences compared to the life sciences, political science and statistics. Using more recent data from the Integrated Postsecondary Education Data System (IPEDS), Bayer and Rouse (2016) find that the share of doctorate degrees in economics awarded to women has stagnated since 1995, with women accounting for approximately 30% of doctorate degrees awarded in 2014. Female representation continues to fall as one progresses up the academic career ladder (Ginther and Kahn, 2004).

2.5 Sensitivity of Estimates to Inclusion and Exclusion of Finance and Econometrics Journals

Finance has emerged as a separate field that coexists with, and sometimes overlaps, with mainstream economics. Given the distinct nature of the field and the existence of separate finance departments in business schools, it is possible that top finance journals are valued differently than other field journals in making tenure decisions. We recognize this reality by conducting separate analyses in Online Appendix Section 5 that test the robustness of our estimates to: (a) pooling economics and finance journals together into combined field journal categories and (b) separating them out. Our results are robust to inclusion or exclusion of finance journals. We similarly test for the sensitivity of estimates to alternative treatments of Econometrics journals in Online Appendix Section 6. Our results are robust to the inclusion and exclusion of Econometrics journals.

3 Junior Faculty Perceptions of Current Tenure and Promotion Practices

We supplement our empirical analysis of job-history and publication data with findings from a survey of individuals currently employed as assistant and associate professors by the top 50 economics departments in the U.S.[49] Respondents were surveyed about their perceptions of how tenure and promotion decisions are determined within their departments, with an emphasis on the role played by T5 publications in these decisions.[50] The survey responses corroborate and contextualize the evidence in Section 2. Junior faculty have rational expectations about the power of the T5. Appendix Section 8.3 presents our survey instrument.

The survey has an overall response rate of 40% (N=308) across all 50 departments, with response rates of 44% (N=210) for assistant professors and 34% (N=97) for associate professors. The overall response rate was highest for departments ranked 41–50 (43%), and lowest for the top 10 departments (37%). Assistant professors had higher response rates than associate professors across all department rank groups except the top 10 departments, for which the response rate was 37% in both groups. Position- and department rank-specific response rates are reported in Online Appendix Figure O-A29.

The response rate gives rise to concerns about non-response bias. Of particular concern is the potential bias that could stem from respondents selecting into the survey based on their ability to publish in the T5. Comparisons of distributions of T5 publications between the respondents and the overall population of assistant and associate professors hired by the top 50 departments provides evidence against this form of selection. Department rank

[49]See Liner and Sewell (2009) for a survey of department chairs on research requirements for promotion and tenure.

[50]The survey was designed with three goals in mind: (i) to confirm our empirical findings about the influence of T5 publications on tenure decisions; (ii) to collect new data on the perceived importance of factors such as teaching performance or external letters that are unobserved in the work-history data; and (iii) to provide junior faculty the opportunity to express their opinions about the consequences (either positive or negative) of current tenure and promotion practices for themselves and for the discipline as a whole.

group-specific Mann-Whitney tests comparing T5 distributions between survey respondents and the overall population fail to reject the null of equality for all rank groups. See Online Appendix Table O-A53 for these comparisons. Online Appendix Section 8.2 presents additional data description for the survey sample.[51]

3.1 Survey Results

One survey question asks respondents to rank eight different areas of research and teaching performance based on their perceptions of the degree to which tenure and/or promotion decisions are influenced by performance in these areas. Figure 11 summarizes responses to this question by presenting the mean ranking assigned by respondents to each performance area. The figure presents three sets of summaries, corresponding to rankings of performance areas for three different types of career advancement: receipt of tenure, promotion to assistant professor, and promotion to associate professor.[52] The quantity of T5 publications receives the highest mean rank across all forms of career advancement. Wilcoxon signed-rank tests performed between pairs of ranking distributions for the eight performance areas indicates that the distribution of rankings of the importance of the quantity of T5 publications is significantly different than the ranking distributions for all of the remaining seven performance areas at the 10% level.[53] In addition to confirming our previous findings of the substantial influence of T5 publications relative to publications in non-T5 journals, these survey results

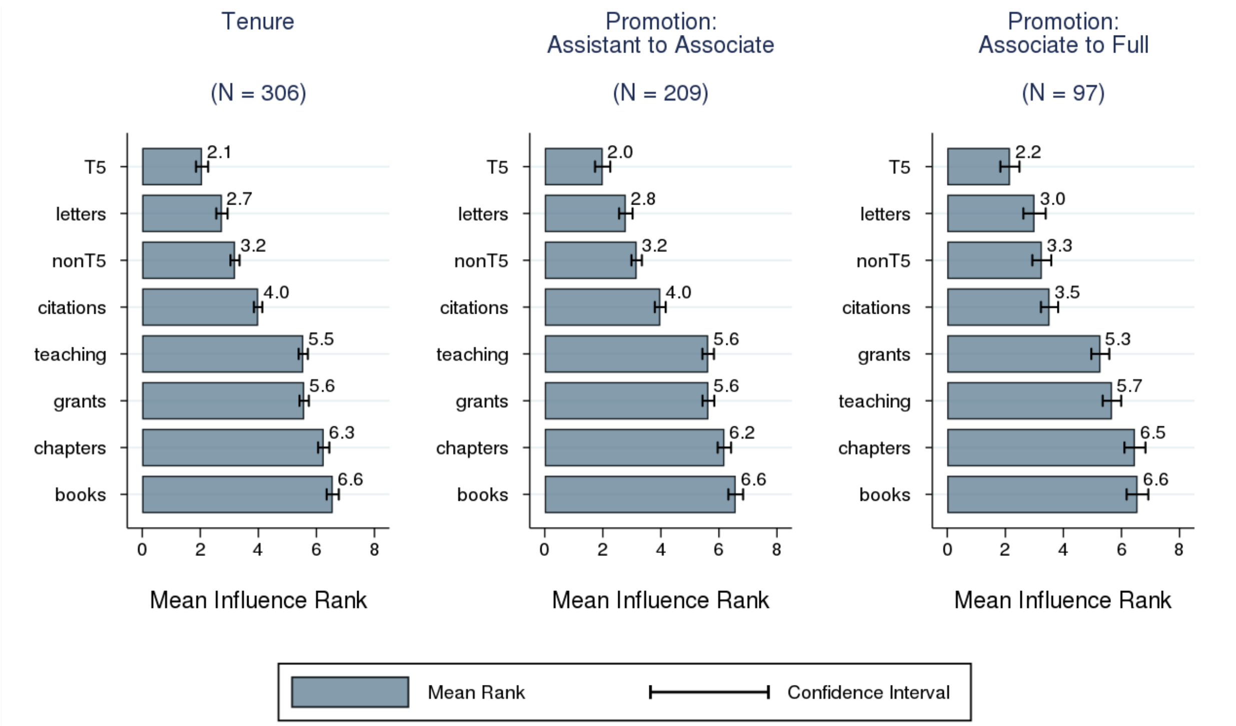
[51]We note that the survey was terminated prematurely because of a complaint to our IRB board by some individuals we attempted to sample. The complainants were concerned that their identity might be determined by our survey protocol despite our efforts to assure anonymity. This source of non-response mechanically leads to a low response rate that does not necessarily produce a bias unless early responders are biased in the same general direction and have axes to grind.

[52]The tenure-specific ranking has a sample size of 306 respondents. The promotion-specific rankings have lower sample sizes because these rankings were presented to different subsets of respondents: rankings for promotion to associate professor was only requested from current assistant professors, and rankings for promotion to full professor was only requested from current associate professors. The reason for employing this form of sample restriction is twofold. First, it ensures that responses are current and well-informed since faculty are only surveyed about promotions to positions that they are currently working towards obtaining. Second, it improves the probability of survey completion by reducing the burden of response for each respondent from 3 to 2 rankings.

[53]See Online Appendix Tables O-A56–O-A58 for pair-wise tests on rankings for each type of career advancement.

reveal that the T5 is also more influential than measures of performance such as external letters of recommendation and teaching performance which are not available to us. These findings support the conclusion that junior faculty at the top departments perceive the quantity of T5 publications to be the most important source of influence on tenure and promotion decisions.

Figure 11: Ranking of Performance Areas Based On Their Perceived Influence On Tenure and Promotion Decisions



Note: This figure summarizes respondents' rankings of 8 performance areas. Responses are summarized by type of career advancement: tenure receipt, promotion to assistant professor, and promotion to associate professor. The bars present mean responses for each performance area. Respondents were given the option to not rank any or all of the eight performance areas. As a result, the number of respondents vary across the performance areas.

The quality of external letters of recommendation receives the second-highest mean ranking across all types of career advancement. External letters are meant to provide tenure and promotion committees an outside view of the quality and impact of candidates' research, especially in comparison to similarly-experienced researchers working in comparable fields. The data do not allow us to test whether one's quantity of T5 publications influences the quality of external letters. However, given that external and internal reviews are both focused